

SQL Code

```
1  /* Create Table/Database*/
2
3  CREATE DATABASE IF NOT EXISTS sales_assignment;
4  -- Query OK, 1 row affected, 1 warning (0.01 sec)
5
6  USE sales_assignment;
7  -- Database changed
8
9
10
11 CREATE TABLE customers (
12     id INT PRIMARY KEY,
13     customer_name VARCHAR(100) NOT NULL,
14     city VARCHAR(100) NOT NULL,
15     region VARCHAR(50) NOT NULL,
16     signup_date DATE NOT NULL
17 );
18 -- Query OK, 0 rows affected (0.03 sec)
19
20 CREATE TABLE products (
21     id INT PRIMARY KEY,
22     product_name VARCHAR(100) NOT NULL,
23     category VARCHAR(50) NOT NULL,
24     unit_price DECIMAL(10,2) NOT NULL
25 );
26 -- Query OK, 0 rows affected (0.03 sec)
27
28 CREATE TABLE orders (
29     id INT PRIMARY KEY,
30     customer_id INT NOT NULL,
31     order_date DATE NOT NULL,
32     region VARCHAR(50) NOT NULL,
33     total_amount DECIMAL(10,2) NOT NULL,
34     FOREIGN KEY (customer_id) REFERENCES customers(id)
35 );
36 -- Query OK, 0 rows affected (0.04 sec)
37
38 CREATE TABLE order_items (
39     id INT PRIMARY KEY,
40     order_id INT NOT NULL,
41     product_id INT NOT NULL,
42     quantity INT NOT NULL,
43     unit_price DECIMAL(10,2) NOT NULL,
44     FOREIGN KEY (order_id) REFERENCES orders(id),
45     FOREIGN KEY (product_id) REFERENCES products(id)
46 );
47 -- Query OK, 0 rows affected (0.04 sec)
48
49 CREATE INDEX idx_orders_customer ON orders(customer_id);
50 -- Query OK, 0 rows affected (0.04 sec)
51 CREATE INDEX idx_orders_date ON orders(order_date);
52 -- Query OK, 0 rows affected (0.02 sec)
53 CREATE INDEX idx_order_items_order ON order_items(order_id);
54 -- Query OK, 0 rows affected (0.02 sec)
55 CREATE INDEX idx_order_items_product ON order_items(product_id);
56 -- Query OK, 0 rows affected (0.03 sec)
57
58 /*Insert Data*/
```

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59 USE sales_assignment;
60 -- Database changed
61
62 INSERT INTO customers (id, customer_name, city, region, signup_date) VALUES
63 (1, 'Aarti Sharma', 'Chicago', 'North', '2023-01-10'),
64 (2, 'Rohit Kumar', 'Dallas', 'South', '2023-02-15'),
65 (3, 'Sneha Gupta', 'Austin', 'West', '2023-03-20'),
66 (4, 'Vikram Patel', 'Boston', 'East', '2023-04-05'),
67 (5, 'Anjali Patel', 'Denver', 'North', '2023-05-12'),
68 (6, 'Amit Singh', 'Phoenix', 'South', '2023-06-01'),
69 (7, 'Priya Reddy', 'Seattle', 'West', '2023-07-14'),
70 (8, 'Siddharth Rao', 'Miami', 'East', '2023-08-22'),
71 (9, 'Kavya Nair', 'Atlanta', 'North', '2023-09-18'),
72 (10, 'Arjun Verma', 'San Diego', 'South', '2023-10-09');
73 -- Query OK, 10 rows affected (0.02 sec)
74
75 INSERT INTO products (id, product_name, category, unit_price) VALUES
76 (1, 'Laptop', 'Electronics', 999.99),
77 (2, 'Wireless Mouse', 'Electronics', 49.99),
78 (3, 'Coffee Maker', 'Home', 79.95),
79 (4, 'Desk Lamp', 'Home', 39.99),
80 (5, 'Running Shoes', 'Sports', 89.99),
81 (6, 'Backpack', 'Sports', 59.99),
82 (7, 'Winter Jacket', 'Fashion', 129.99),
83 (8, 'Leather Wallet', 'Fashion', 49.99);
84 -- Query OK, 8 rows affected (0.01 sec)
85
86 INSERT INTO orders (id, customer_id, order_date, region, total_amount) VALUES
87 (1, 1, '2024-01-05', 'North', 1049.98),
88 (2, 2, '2024-01-12', 'South', 249.95),
89 (3, 1, '2024-02-03', 'North', 89.98),
90 (4, 3, '2024-02-18', 'West', 299.97),
91 (5, 4, '2024-03-07', 'East', 159.98),
92 (6, 5, '2024-03-20', 'North', 149.95),
93 (7, 6, '2024-04-02', 'South', 279.90),
94 (8, 7, '2024-04-12', 'West', 349.95),
95 (9, 8, '2024-05-01', 'East', 199.90),
96 (10, 9, '2024-05-15', 'North', 239.85),
97 (11, 10, '2024-06-01', 'South', 129.95),
98 (12, 2, '2024-06-20', 'South', 89.98);
99 -- Query OK, 12 rows affected (0.01 sec)
100
101 INSERT INTO order_items (id, order_id, product_id, quantity, unit_price) VALUES
102 (1, 1, 1, 1, 999.99),
103 (2, 1, 2, 1, 49.99),
104 (3, 2, 3, 1, 79.95),
105 (4, 2, 4, 1, 39.99),
106 (5, 2, 5, 1, 129.99),
107 (6, 3, 6, 1, 59.99),
108 (7, 4, 7, 1, 129.99),
109 (8, 4, 8, 1, 49.99),
110 (9, 4, 5, 1, 89.99),
111 (10, 5, 4, 1, 39.99),
112 (11, 5, 6, 1, 59.99),
113 (12, 6, 5, 1, 89.99),
114 (13, 7, 3, 2, 79.95),
115 (14, 8, 7, 1, 129.99),
116 (15, 8, 8, 1, 49.99),
117 (16, 9, 2, 2, 49.99),
118 (17, 10, 1, 1, 999.99),
119 (18, 10, 4, 1, 39.99),
120 (19, 11, 6, 1, 59.99),

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121 (20, 12, 3, 1, 79.95);
122 -- Query OK, 20 rows affected (0.02 sec)
123
124
125 /* Section_A_basic_queries.sql */
126
127 USE sales_assignment;
128
129 SELECT table_name
130 FROM information_schema.tables
131 WHERE table_schema = 'sales_assignment'
132 ORDER BY table_name;
133
134 SELECT * FROM customers ORDER BY id LIMIT 5;
135 SELECT * FROM products ORDER BY id LIMIT 5;
136 SELECT * FROM orders ORDER BY id LIMIT 5;
137 SELECT * FROM order_items ORDER BY id LIMIT 5;
138
139 SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM customers
140 UNION ALL
141 SELECT 'products', COUNT(*) FROM products
142 UNION ALL
143 SELECT 'orders', COUNT(*) FROM orders
144 UNION ALL
145 SELECT 'order_items', COUNT(*) FROM order_items;
146
147 /* Setup
148
149 mysql> CREATE DATABASE IF NOT EXISTS sales_assignment;
150 Query OK, 1 row affected, 1 warning (0.01 sec)
151
152 mysql> USE sales_assignment;
153 Database changed
154 mysql> CREATE TABLE customers (
155     ->     id INT PRIMARY KEY,
156     ->     customer_name VARCHAR(100) NOT NULL,
157     ->     city VARCHAR(100) NOT NULL,
158     ->     region VARCHAR(50) NOT NULL,
159     ->     signup_date DATE NOT NULL
160     -> );
161 Query OK, 0 rows affected (0.03 sec)
162
163 mysql> CREATE TABLE products (
164     ->     id INT PRIMARY KEY,
165     ->     product_name VARCHAR(100) NOT NULL,
166     ->     category VARCHAR(50) NOT NULL,
167     ->     unit_price DECIMAL(10,2) NOT NULL
168     -> );
169 Query OK, 0 rows affected (0.03 sec)
170
171 mysql> CREATE TABLE orders (
172     ->     id INT PRIMARY KEY,
173     ->     customer_id INT NOT NULL,
174     ->     order_date DATE NOT NULL,
175     ->     region VARCHAR(50) NOT NULL,
176     ->     total_amount DECIMAL(10,2) NOT NULL,
177     ->     FOREIGN KEY (customer_id) REFERENCES customers(id)
178     -> );
179 Query OK, 0 rows affected (0.04 sec)
180
181 mysql> CREATE TABLE order_items (
182     ->     id INT PRIMARY KEY,

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183     -> order_id INT NOT NULL,
184     -> product_id INT NOT NULL,
185     -> quantity INT NOT NULL,
186     -> unit_price DECIMAL(10,2) NOT NULL,
187     -> FOREIGN KEY (order_id) REFERENCES orders(id),
188     -> FOREIGN KEY (product_id) REFERENCES products(id)
189     -> );
190 Query OK, 0 rows affected (0.04 sec)
191
192 mysql> CREATE INDEX idx_orders_customer ON orders(customer_id);
193 Query OK, 0 rows affected (0.04 sec)
194 Records: 0 Duplicates: 0 Warnings: 0
195
196 mysql> CREATE INDEX idx_orders_date ON orders(order_date);
197 Query OK, 0 rows affected (0.02 sec)
198 Records: 0 Duplicates: 0 Warnings: 0
199
200 mysql> CREATE INDEX idx_order_items_order ON order_items(order_id);
201 Query OK, 0 rows affected (0.02 sec)
202 Records: 0 Duplicates: 0 Warnings: 0
203
204 mysql> CREATE INDEX idx_order_items_product ON order_items(product_id);
205 Query OK, 0 rows affected (0.03 sec)
206 Records: 0 Duplicates: 0 Warnings: 0
207
208 Section A
209
210 mysql> USE sales_assignment;
211 Database changed
212 mysql> INSERT INTO customers (id, customer_name, city, region, signup_date) VALUES
213     -> (1, 'Aarti Sharma', 'Chicago', 'North', '2023-01-10'),
214     -> (2, 'Rohit Kumar', 'Dallas', 'South', '2023-02-15'),
215     -> (3, 'Sneha Gupta', 'Austin', 'West', '2023-03-20'),
216     -> (4, 'Vikram Patel', 'Boston', 'East', '2023-04-05'),
217     -> (5, 'Anjali Patel', 'Denver', 'North', '2023-05-12'),
218     -> (6, 'Amit Singh', 'Phoenix', 'South', '2023-06-01'),
219     -> (7, 'Priya Reddy', 'Seattle', 'West', '2023-07-14'),
220     -> (8, 'Siddharth Rao', 'Miami', 'East', '2023-08-22'),
221     -> (9, 'Kavya Nair', 'Atlanta', 'North', '2023-09-18'),
222     -> (10, 'Arjun Verma', 'San Diego', 'South', '2023-10-09');
223 Query OK, 10 rows affected (0.02 sec)
224 Records: 10 Duplicates: 0 Warnings: 0
225
226 mysql> INSERT INTO products (id, product_name, category, unit_price) VALUES
227     -> (1, 'Laptop', 'Electronics', 999.99),
228     -> (2, 'Wireless Mouse', 'Electronics', 49.99),
229     -> (3, 'Coffee Maker', 'Home', 79.95),
230     -> (4, 'Desk Lamp', 'Home', 39.99),
231     -> (5, 'Running Shoes', 'Sports', 89.99),
232     -> (6, 'Backpack', 'Sports', 59.99),
233     -> (7, 'Winter Jacket', 'Fashion', 129.99),
234     -> (8, 'Leather Wallet', 'Fashion', 49.99);
235 Query OK, 8 rows affected (0.01 sec)
236 Records: 8 Duplicates: 0 Warnings: 0
237
238 mysql> INSERT INTO orders (id, customer_id, order_date, region, total_amount) VALUES
239     -> (1, 1, '2024-01-05', 'North', 1049.98),
240     -> (2, 2, '2024-01-12', 'South', 249.95),
241     -> (3, 1, '2024-02-03', 'North', 89.98),
242     -> (4, 3, '2024-02-18', 'West', 299.97),
243     -> (5, 4, '2024-03-07', 'East', 159.98),
244     -> (6, 5, '2024-03-20', 'North', 149.95),

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245     -> (7, 6, '2024-04-02', 'South', 279.90),
246     -> (8, 7, '2024-04-12', 'West', 349.95),
247     -> (9, 8, '2024-05-01', 'East', 199.90),
248     -> (10, 9, '2024-05-15', 'North', 239.85),
249     -> (11, 10, '2024-06-01', 'South', 129.95),
250     -> (12, 2, '2024-06-20', 'South', 89.98);
251 Query OK, 12 rows affected (0.01 sec)
252 Records: 12 Duplicates: 0 Warnings: 0
253
254 mysql> INSERT INTO order_items (id, order_id, product_id, quantity, unit_price) VALUES
255     -> (1, 1, 1, 1, 999.99),
256     -> (2, 1, 2, 1, 49.99),
257     -> (3, 2, 3, 1, 79.95),
258     -> (4, 2, 4, 1, 39.99),
259     -> (5, 2, 5, 1, 129.99),
260     -> (6, 3, 6, 1, 59.99),
261     -> (7, 4, 7, 1, 129.99),
262     -> (8, 4, 8, 1, 49.99),
263     -> (9, 4, 5, 1, 89.99),
264     -> (10, 5, 4, 1, 39.99),
265     -> (11, 5, 6, 1, 59.99),
266     -> (12, 6, 5, 1, 89.99),
267     -> (13, 7, 3, 2, 79.95),
268     -> (14, 8, 7, 1, 129.99),
269     -> (15, 8, 8, 1, 49.99),
270     -> (16, 9, 2, 2, 49.99),
271     -> (17, 10, 1, 1, 999.99),
272     -> (18, 10, 4, 1, 39.99),
273     -> (19, 11, 6, 1, 59.99),
274     -> (20, 12, 3, 1, 79.95);
275 Query OK, 20 rows affected (0.02 sec)
276 Records: 20 Duplicates: 0 Warnings: 0
277
278 mysql> USE sales_assignment;
279 Database changed
280 mysql>
281 mysql> SELECT table_name
282     -> FROM information_schema.tables
283     -> WHERE table_schema = 'sales_assignment'
284     -> ORDER BY table_name;
285 +-----+
286 | TABLE_NAME |
287 +-----+
288 | customers   |
289 | order_items |
290 | orders      |
291 | products    |
292 +-----+
293 4 rows in set (0.02 sec)
294
295 mysql> SELECT * FROM customers ORDER BY id LIMIT 5;
296 +---+-----+-----+-----+-----+
297 | id | customer_name | city    | region | signup_date |
298 +---+-----+-----+-----+-----+
299 | 1  | Aarti Sharma  | Chicago | North  | 2023-01-10  |
300 | 2  | Rohit Kumar   | Dallas  | South  | 2023-02-15  |
301 | 3  | Sneha Gupta   | Austin  | West   | 2023-03-20  |
302 | 4  | Vikram Patel  | Boston  | East   | 2023-04-05  |
303 | 5  | Anjali Patel  | Denver  | North  | 2023-05-12  |
304 +---+-----+-----+-----+-----+
305 5 rows in set (0.00 sec)
306

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307 mysql> SELECT * FROM products ORDER BY id LIMIT 5;
308 +-----+-----+-----+-----+
309 | id | product_name | category | unit_price |
310 +-----+-----+-----+-----+
311 | 1 | Laptop | Electronics | 999.99 |
312 | 2 | Wireless Mouse | Electronics | 49.99 |
313 | 3 | Coffee Maker | Home | 79.95 |
314 | 4 | Desk Lamp | Home | 39.99 |
315 | 5 | Running Shoes | Sports | 89.99 |
316 +-----+-----+-----+-----+
317 5 rows in set (0.00 sec)
318
319 mysql> SELECT * FROM orders ORDER BY id LIMIT 5;
320 +-----+-----+-----+-----+-----+
321 | id | customer_id | order_date | region | total_amount |
322 +-----+-----+-----+-----+-----+
323 | 1 | 1 | 2024-01-05 | North | 1049.98 |
324 | 2 | 2 | 2024-01-12 | South | 249.95 |
325 | 3 | 1 | 2024-02-03 | North | 89.98 |
326 | 4 | 3 | 2024-02-18 | West | 299.97 |
327 | 5 | 4 | 2024-03-07 | East | 159.98 |
328 +-----+-----+-----+-----+-----+
329 5 rows in set (0.00 sec)
330
331 mysql> SELECT * FROM order_items ORDER BY id LIMIT 5;
332 +-----+-----+-----+-----+-----+
333 | id | order_id | product_id | quantity | unit_price |
334 +-----+-----+-----+-----+-----+
335 | 1 | 1 | 1 | 1 | 999.99 |
336 | 2 | 1 | 2 | 1 | 49.99 |
337 | 3 | 2 | 3 | 1 | 79.95 |
338 | 4 | 2 | 4 | 1 | 39.99 |
339 | 5 | 2 | 5 | 1 | 129.99 |
340 +-----+-----+-----+-----+-----+
341 5 rows in set (0.00 sec)
342
343 mysql> SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM customers
344 -> UNION ALL
345 -> SELECT 'products', COUNT(*) FROM products
346 -> UNION ALL
347 -> SELECT 'orders', COUNT(*) FROM orders
348 -> UNION ALL
349 -> SELECT 'order_items', COUNT(*) FROM order_items;
350 +-----+-----+
351 | table_name | row_count |
352 +-----+-----+
353 | customers | 10 |
354 | products | 8 |
355 | orders | 12 |
356 | order_items | 20 |
357 +-----+-----+
358 4 rows in set (0.01 sec)
359
360 */
361
362
363 /* Section_B_filtering_queries.sql */
364
365 USE sales_assignment;
366
367 SELECT *
368 FROM orders

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369 WHERE region = 'North'
370 ORDER BY order_date;
371
372 SELECT *
373 FROM products
374 WHERE category = 'Electronics' AND unit_price > 50
375 ORDER BY unit_price DESC;
376
377 SELECT *
378 FROM orders
379 WHERE total_amount > 200
380 ORDER BY total_amount DESC;
381
382 SELECT *
383 FROM customers
384 WHERE signup_date < '2023-06-01'
385 ORDER BY signup_date;
386
387
388 /* Section B
389
390 mysql> USE sales_assignment;
391 Database changed
392 mysql> SELECT *
393     -> FROM orders
394     -> WHERE region = 'North'
395     -> ORDER BY order_date;
396 +-----+-----+-----+-----+-----+
397 | id | customer_id | order_date | region | total_amount |
398 +-----+-----+-----+-----+-----+
399 | 1 | 1 | 2024-01-05 | North | 1049.98 |
400 | 3 | 1 | 2024-02-03 | North | 89.98 |
401 | 6 | 5 | 2024-03-20 | North | 149.95 |
402 | 10 | 9 | 2024-05-15 | North | 239.85 |
403 +-----+-----+-----+-----+-----+
404 4 rows in set (0.00 sec)
405
406 mysql> SELECT *
407     -> FROM products
408     -> WHERE category = 'Electronics' AND unit_price > 50
409     -> ORDER BY unit_price DESC;
410 +-----+-----+-----+-----+
411 | id | product_name | category | unit_price |
412 +-----+-----+-----+-----+
413 | 1 | Laptop | Electronics | 999.99 |
414 +-----+-----+-----+-----+
415 1 row in set (0.00 sec)
416
417 mysql> SELECT *
418     -> FROM orders
419     -> WHERE total_amount > 200
420     -> ORDER BY total_amount DESC;
421 +-----+-----+-----+-----+-----+
422 | id | customer_id | order_date | region | total_amount |
423 +-----+-----+-----+-----+-----+
424 | 1 | 1 | 2024-01-05 | North | 1049.98 |
425 | 8 | 7 | 2024-04-12 | West | 349.95 |
426 | 4 | 3 | 2024-02-18 | West | 299.97 |
427 | 7 | 6 | 2024-04-02 | South | 279.90 |
428 | 2 | 2 | 2024-01-12 | South | 249.95 |
429 | 10 | 9 | 2024-05-15 | North | 239.85 |
430 +-----+-----+-----+-----+-----+

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```

431 6 rows in set (0.00 sec)
432
433 mysql> SELECT *
434     -> FROM customers
435     -> WHERE signup_date < '2023-06-01'
436     -> ORDER BY signup_date;
437 +-----+-----+-----+-----+-----+
438 | id | customer_name | city   | region | signup_date |
439 +-----+-----+-----+-----+-----+
440 |  1 | Aarti Sharma  | Chicago | North  | 2023-01-10 |
441 |  2 | Rohit Kumar   | Dallas  | South  | 2023-02-15 |
442 |  3 | Sneha Gupta   | Austin  | West   | 2023-03-20 |
443 |  4 | Vikram Patel  | Boston  | East   | 2023-04-05 |
444 |  5 | Anjali Patel  | Denver  | North  | 2023-05-12 |
445 +-----+-----+-----+-----+-----+
446 5 rows in set (0.00 sec)
447
448
449 */
450
451 /* Section_C_aggregation_queries.sql */
452
453 USE sales_assignment;
454
455 SELECT region, SUM(total_amount) AS total_sales
456 FROM orders
457 GROUP BY region
458 ORDER BY total_sales DESC;
459
460 SELECT p.product_name, SUM(oi.quantity) AS total_quantity
461 FROM order_items oi
462 JOIN products p ON oi.product_id = p.id
463 GROUP BY p.product_name
464 ORDER BY total_quantity DESC;
465
466 SELECT c.customer_name, AVG(o.total_amount) AS avg_order_value
467 FROM orders o
468 JOIN customers c ON o.customer_id = c.id
469 GROUP BY c.customer_name
470 ORDER BY avg_order_value DESC;
471
472 SELECT category, MAX(unit_price) AS max_unit_price
473 FROM products
474 GROUP BY category;
475
476 /*
477 Section C
478
479
480 mysql> USE sales_assignment;
481 Database changed
482 mysql> SELECT region, SUM(total_amount) AS total_sales
483     -> FROM orders
484     -> GROUP BY region
485     -> ORDER BY total_sales DESC;
486 +-----+-----+
487 | region | total_sales |
488 +-----+-----+
489 | North  | 1529.76 |
490 | South  | 749.78 |
491 | West   | 649.92 |
492 | East   | 359.88 |

```



```

493 +-----+
494 4 rows in set (0.00 sec)
495
496 mysql>
497 mysql> SELECT p.product_name, SUM(oi.quantity) AS total_quantity
498     -> FROM order_items oi
499     -> JOIN products p ON oi.product_id = p.id
500     -> GROUP BY p.product_name
501     -> ORDER BY total_quantity DESC;
502 +-----+-----+
503 | product_name | total_quantity |
504 +-----+-----+
505 | Coffee Maker |             4 |
506 | Wireless Mouse |            3 |
507 | Desk Lamp |            3 |
508 | Running Shoes |            3 |
509 | Backpack |            3 |
510 | Laptop |            2 |
511 | Winter Jacket |            2 |
512 | Leather Wallet |            2 |
513 +-----+-----+
514 8 rows in set (0.00 sec)
515
516 mysql>
517 mysql> SELECT c.customer_name, AVG(o.total_amount) AS avg_order_value
518     -> FROM orders o
519     -> JOIN customers c ON o.customer_id = c.id
520     -> GROUP BY c.customer_name
521     -> ORDER BY avg_order_value DESC;
522 +-----+-----+
523 | customer_name | avg_order_value |
524 +-----+-----+
525 | Aarti Sharma |    569.980000 |
526 | Priya Reddy |    349.950000 |
527 | Sneha Gupta |    299.970000 |
528 | Amit Singh |    279.900000 |
529 | Kavya Nair |    239.850000 |
530 | Siddharth Rao |    199.900000 |
531 | Rohit Kumar |    169.965000 |
532 | Vikram Patel |    159.980000 |
533 | Anjali Patel |    149.950000 |
534 | Arjun Verma |    129.950000 |
535 +-----+-----+
536 10 rows in set (0.00 sec)
537
538 mysql>
539 mysql> SELECT category, MAX(unit_price) AS max_unit_price
540     -> FROM products
541     -> GROUP BY category;
542 +-----+-----+
543 | category | max_unit_price |
544 +-----+-----+
545 | Electronics |         999.99 |
546 | Home |          79.95 |
547 | Sports |          89.99 |
548 | Fashion |         129.99 |
549 +-----+-----+
550 4 rows in set (0.00 sec)
551
552
553 */
554

```

```

555  /* Section_D_joins_queries.sql */
556
557  USE sales_assignment;
558
559  SELECT o.id, c.customer_name, o.order_date, o.region, o.total_amount
560  FROM orders o
561  JOIN customers c ON o.customer_id = c.id
562  ORDER BY o.id;
563
564  SELECT oi.id, o.id AS order_id, p.product_name, oi.quantity, oi.unit_price
565  FROM order_items oi
566  JOIN orders o ON oi.order_id = o.id
567  JOIN products p ON oi.product_id = p.id
568  ORDER BY oi.id;
569
570  SELECT c.customer_name, SUM(o.total_amount) AS total_spend
571  FROM customers c
572  LEFT JOIN orders o ON c.id = o.customer_id
573  GROUP BY c.customer_name
574  ORDER BY total_spend DESC;
575
576
577  /*
578  Section D
579
580
581  mysql> USE sales_assignment;
582  Database changed
583  mysql>
584  mysql> SELECT o.id, c.customer_name, o.order_date, o.region, o.total_amount
585         -> FROM orders o
586         -> JOIN customers c ON o.customer_id = c.id
587         -> ORDER BY o.id;
588
589  +----+-----+-----+-----+-----+
590  | id | customer_name | order_date | region | total_amount |
591  +----+-----+-----+-----+-----+
592  | 1 | Aarti Sharma | 2024-01-05 | North | 1049.98 |
593  | 2 | Rohit Kumar | 2024-01-12 | South | 249.95 |
594  | 3 | Aarti Sharma | 2024-02-03 | North | 89.98 |
595  | 4 | Sneha Gupta | 2024-02-18 | West | 299.97 |
596  | 5 | Vikram Patel | 2024-03-07 | East | 159.98 |
597  | 6 | Anjali Patel | 2024-03-20 | North | 149.95 |
598  | 7 | Amit Singh | 2024-04-02 | South | 279.90 |
599  | 8 | Priya Reddy | 2024-04-12 | West | 349.95 |
600  | 9 | Siddharth Rao | 2024-05-01 | East | 199.90 |
601  | 10 | Kavya Nair | 2024-05-15 | North | 239.85 |
602  | 11 | Arjun Verma | 2024-06-01 | South | 129.95 |
603  | 12 | Rohit Kumar | 2024-06-20 | South | 89.98 |
604  +----+-----+-----+-----+-----+
605  12 rows in set (0.00 sec)
606
607  mysql>
608  mysql> SELECT oi.id, o.id AS order_id, p.product_name, oi.quantity, oi.unit_price
609         -> FROM order_items oi
610         -> JOIN orders o ON oi.order_id = o.id
611         -> JOIN products p ON oi.product_id = p.id
612         -> ORDER BY oi.id;
613
614  +----+-----+-----+-----+-----+
615  | id | order_id | product_name | quantity | unit_price |
616  +----+-----+-----+-----+-----+
617  | 1 | 1 | Laptop | 1 | 999.99 |
618  | 2 | 1 | Wireless Mouse | 1 | 49.99 |

```

617	3	2	Coffee Maker	1	79.95
618	4	2	Desk Lamp	1	39.99
619	5	2	Running Shoes	1	129.99
620	6	3	Backpack	1	59.99
621	7	4	Winter Jacket	1	129.99
622	8	4	Leather Wallet	1	49.99
623	9	4	Running Shoes	1	89.99
624	10	5	Desk Lamp	1	39.99
625	11	5	Backpack	1	59.99
626	12	6	Running Shoes	1	89.99
627	13	7	Coffee Maker	2	79.95
628	14	8	Winter Jacket	1	129.99
629	15	8	Leather Wallet	1	49.99
630	16	9	Wireless Mouse	2	49.99
631	17	10	Laptop	1	999.99
632	18	10	Desk Lamp	1	39.99
633	19	11	Backpack	1	59.99
634	20	12	Coffee Maker	1	79.95

```
635 +-----+-----+-----+-----+-----+-----+
```

```
636 20 rows in set (0.00 sec)
```

```
637
638 mysql>
639 mysql> SELECT c.customer_name, SUM(o.total_amount) AS total_spend
640     -> FROM customers c
641     -> LEFT JOIN orders o ON c.id = o.customer_id
642     -> GROUP BY c.customer_name
643     -> ORDER BY total_spend DESC;
```

customer_name	total_spend
Aarti Sharma	1139.96
Priya Reddy	349.95
Rohit Kumar	339.93
Sneha Gupta	299.97
Amit Singh	279.90
Kavya Nair	239.85
Siddharth Rao	199.90
Vikram Patel	159.98
Anjali Patel	149.95
Arjun Verma	129.95

```
657 +-----+-----+-----+-----+-----+-----+
```

```
658 10 rows in set (0.00 sec)
```

```
659
660
661 */
662
663 /* Section E_subqueries_queries.sql */
664
665 USE sales_assignment;
666
667 SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(total_amount) AS monthly_sales
668 FROM orders
669 GROUP BY DATE_FORMAT(order_date, '%Y-%m')
670 ORDER BY month;
671
672 SELECT c.customer_name, SUM(o.total_amount) AS total_spend
673 FROM customers c
674 JOIN orders o ON c.id = o.customer_id
675 GROUP BY c.customer_name
676 ORDER BY total_spend DESC
677 LIMIT 3;
```

```
678
```

```

679 SELECT customer_id, order_date, COUNT(*) AS duplicate_count
680 FROM orders
681 GROUP BY customer_id, order_date
682 HAVING COUNT(*) > 1;
683
684 SELECT id, total_amount,
685        CASE
686            WHEN total_amount >= 300 THEN 'High Value'
687            WHEN total_amount >= 150 THEN 'Medium Value'
688            ELSE 'Low Value'
689        END AS value_category
690 FROM orders
691 ORDER BY total_amount DESC;
692
693 /*
694 Section E
695
696 mysql> USE sales_assignment;
697 Database changed
698 mysql>
699 mysql> SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(total_amount) AS monthly_sales
700        -> FROM orders
701        -> GROUP BY DATE_FORMAT(order_date, '%Y-%m')
702        -> ORDER BY month;
703 +-----+-----+
704 | month | monthly_sales |
705 +-----+-----+
706 | 2024-01 | 1299.93 |
707 | 2024-02 | 389.95 |
708 | 2024-03 | 309.93 |
709 | 2024-04 | 629.85 |
710 | 2024-05 | 439.75 |
711 | 2024-06 | 219.93 |
712 +-----+-----+
713 6 rows in set (0.00 sec)
714
715 mysql>
716 mysql> SELECT c.customer_name, SUM(o.total_amount) AS total_spend
717        -> FROM customers c
718        -> JOIN orders o ON c.id = o.customer_id
719        -> GROUP BY c.customer_name
720        -> ORDER BY total_spend DESC
721        -> LIMIT 3;
722 +-----+-----+
723 | customer_name | total_spend |
724 +-----+-----+
725 | Aarti Sharma | 1139.96 |
726 | Priya Reddy | 349.95 |
727 | Rohit Kumar | 339.93 |
728 +-----+-----+
729 3 rows in set (0.00 sec)
730
731 mysql>
732 mysql> SELECT customer_id, order_date, COUNT(*) AS duplicate_count
733        -> FROM orders
734        -> GROUP BY customer_id, order_date
735        -> HAVING COUNT(*) > 1;
736 Empty set (0.00 sec)
737
738 mysql>
739 mysql> SELECT id, total_amount,
740        -> CASE

```

```

741     ->         WHEN total_amount >= 300 THEN 'High Value'
742     ->         WHEN total_amount >= 150 THEN 'Medium Value'
743     ->         ELSE 'Low Value'
744     ->     END AS value_category
745     -> FROM orders
746     -> ORDER BY total_amount DESC;
747 +-----+-----+-----+
748 | id | total_amount | value_category |
749 +-----+-----+-----+
750 |  1 |    1049.98 | High Value    |
751 |  8 |     349.95 | High Value    |
752 |  4 |     299.97 | Medium Value  |
753 |  7 |     279.90 | Medium Value  |
754 |  2 |     249.95 | Medium Value  |
755 | 10 |     239.85 | Medium Value  |
756 |  9 |     199.90 | Medium Value  |
757 |  5 |     159.98 | Medium Value  |
758 |  6 |     149.95 | Low Value     |
759 | 11 |     129.95 | Low Value     |
760 |  3 |      89.98 | Low Value     |
761 | 12 |      89.98 | Low Value     |
762 +-----+-----+-----+
763 12 rows in set (0.00 sec)
764 */

```